

Advanced Stochastic Processes Midterm Exam

Spring 2026

This exam contains 6 problems and is worth 100 points in total. Provide complete proofs and necessary calculations.

1. (10 points) Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and let $B \in \mathcal{F}$.

(a) Define

$$\mathcal{G}_B = \{A \cap B : A \in \mathcal{F}\}.$$

Prove that $\mathcal{G}_B$ is a σ -field on the set B .

- (b) Let X be a nonnegative integer-valued random variable. Prove that

$$\mathbb{E}[X] = \sum_{k=1}^{\infty} \mathbb{P}(X \geq k) = \sum_{k=0}^{\infty} \mathbb{P}(X > k).$$

2. (10 points) Let $\{X_n\}_{n \geq 0}$ and $\{Y_n\}_{n \geq 0}$ be two independent ergodic discrete-time Markov chains with state spaces S_X and S_Y , respectively. Their one-step transition probabilities are

$$p(x, x') = \mathbb{P}(X_{n+1} = x' \mid X_n = x), \quad q(y, y') = \mathbb{P}(Y_{n+1} = y' \mid Y_n = y),$$

and their stationary distributions are π and ν , respectively.

Define a new process by

$$Z_n = (X_n, Y_n), \quad n \geq 0.$$

Prove that $\{Z_n\}_{n \geq 0}$ is also an ergodic Markov chain, and determine:

- (a) the one-step transition probabilities of Z_n ;
(b) a stationary distribution of Z_n .
3. (16 points) Let $\{X_n\}_{n \geq 0}$ be a stochastic process on a countable state space S . Prove that the following two conditions are equivalent:

- (i) For every $n \geq 0$ and every $i_0, \dots, i_n, j \in S$, whenever the conditional probability is well defined,

$$\mathbb{P}(X_{n+1} = j \mid X_0 = i_0, \dots, X_n = i_n) = \mathbb{P}(X_{n+1} = j \mid X_n = i_n).$$

- (ii) For every $m \geq 1$, $n \geq 0$, and every $i_0, \dots, i_n, j \in S$, whenever the conditional probability is well defined,

$$\mathbb{P}(X_{n+m} = j \mid X_0 = i_0, \dots, X_n = i_n) = \mathbb{P}(X_{n+m} = j \mid X_n = i_n).$$

In other words, prove that the Markov property is equivalent to the statement that the conditional distribution of any future state X_{n+m} depends only on the present state X_n and not on the earlier history.

4. (20 points) Consider a Markov chain on the state space $S = \{0, 1, 2\}$ with transition matrix

$$P = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{bmatrix}.$$

- (a) Prove that every state of this chain is recurrent.
- (b) Find the mean recurrence time of each state.
5. (30 points) Let $\{N_1(t)\}_{t \geq 0}$ and $\{N_2(t)\}_{t \geq 0}$ be two independent Poisson processes with rates $\lambda > 0$ and $\mu > 0$, respectively. Define

$$N(t) = N_1(t) + N_2(t), \quad t \geq 0.$$

- (a) Prove that $\{N(t)\}_{t \geq 0}$ is again a Poisson process, and determine its rate.
- (b) Prove that the probability that the first event of $N(t)$ comes from $N_1(t)$ is

$$\frac{\lambda}{\lambda + \mu}.$$

6. (10 points) Let $\{X(t)\}_{t \geq 0}$ and $\{Y(t)\}_{t \geq 0}$ be two independent Poisson processes such that

$$\mathbb{E}[X(t)] = \lambda t, \quad \mathbb{E}[Y(t)] = \mu t.$$

Let T and T' be two successive event times of the process $X(t)$, with $T' < \infty$, such that

$$T < T', \quad X(t) = X(T) \text{ for } T < t < T', \quad X(T') = X(T) + 1.$$

Define

$$N = Y(T') - Y(T),$$

that is, N is the number of events of the process $Y(t)$ occurring during the time interval (T, T') .

Find the distribution of N .